Complex Numbers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Basic arithmetic with complex numbers**

Given $z_1 = 4 + 3i$ and $z_2 = 2 - 5i$, find $z_1 + z_2$ and $z_1 - z_2$.

$z_1 + z_2 = (4 + 2) + (3 - 5)i = 6 - 2i$. $z_1 - z_2 = (4 - 2) + (3 - (-5))i = 2 + 8i$.

2 **[Foundational] • Target: Basic arithmetic with complex numbers**

Find $z_1 z_2$ for $z_1 = 2 + 3i$ and $z_2 = 4 - i$.

$z_1 z_2 = (2 + 3i)(4 - i) = 8 - 2i + 12i - 3i^2 = 8 + 10i + 3 = 11 + 10i$.

3 **[Foundational] • Target: The complex conjugate and division**

Find the conjugate of $z = 7 - 2i$, and compute $z\bar{z}$.

$\bar{z} = 7 + 2i$. $z\bar{z} = (7 - 2i)(7 + 2i) = 49 - 4i^2 = 49 + 4 = 53$.

4 **[Foundational] • Target: The complex conjugate and division**

Simplify $\frac{3 + i}{2 - 3i}$ by multiplying by the conjugate of the denominator.

$\frac{3+i}{2-3i} \cdot \frac{2+3i}{2+3i} = \frac{(3+i)(2+3i)}{4+9} = \frac{6+9i+2i+3i^2}{13} = \frac{3+11i}{13} = \frac{3}{13} + \frac{11}{13}i$.

5 **[Foundational] • Target: Modulus and the Argand diagram**

Find the modulus $|z|$ of $z = 5 - 12i$.

$|z| = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

6 **[Foundational] • Target: Modulus and the Argand diagram**

The graph shows several complex numbers plotted on the Argand diagram (real part horizontal, imaginary part vertical).

a Read off the complex number plotted at $(4, 1)$ in Cartesian form. The point $(4, 1)$ corresponds to $4 + i$.

b Find the modulus of the complex number plotted at $(-2, -3)$. The point $(-2, -3)$ corresponds to $-2 - 3i$, with modulus $\sqrt{(-2)^2 + (-3)^2} = \sqrt{13}$.

(a) The point $(4, 1)$ corresponds to $4 + i$.; (b) The point $(-2, -3)$ corresponds to $-2 - 3i$, with modulus $\sqrt{(-2)^2 + (-3)^2} = \sqrt{13}$.

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7 [Foundational] • Target: Solving equations with complex roots

Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.

By the quadratic formula: $x = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving equations with complex roots

One root of a quadratic equation with real coefficients is $3 - 4i$. State the other root, and find the quadratic equation.

Since the coefficients are real, complex roots occur in conjugate pairs, so the other root is $3 + 4i$. Sum of roots $= 6$; product $= (3 - 4i)(3 + 4i) = 9 + 16 = 25$. Equation: $x^2 - 6x + 25 = 0$.

9 [Intermediate] • Target: Solving equations with complex roots

Given that $z = 1 + 2i$ is a root of $z^3 - 4z^2 + 9z - 10 = 0$ (with real coefficients), find the other two roots.

Since the coefficients are real, $z = 1 - 2i$ is also a root. These two roots multiply to $(1 + 2i)(1 - 2i) = 1 + 4 = 5$ and sum to 2, giving the quadratic factor $z^2 - 2z + 5$. Dividing the cubic by this factor: $z^3 - 4z^2 + 9z - 10 \div (z^2 - 2z + 5) = z - 2$, so the third root is $z = 2$. The other two roots are $1 - 2i$ and 2.

10 [Intermediate] • Target: Converting to polar (modulus-argument) form

Write $z = \sqrt{3} + i$ in polar form $r(\cos\theta + i\sin\theta)$, giving the argument in radians.

$r = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$. $\theta = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$. Polar form: $z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$.

11 [Intermediate] • Target: Converting to polar (modulus-argument) form

Convert $z = 6\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ to Cartesian form.

$a = 6\cos\left(\frac{2\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$. $b = 6\sin\left(\frac{2\pi}{3}\right) = 6\left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$. So $z = -3 + 3\sqrt{3}i$.

12 [Intermediate] • Target: Multiplying and dividing in polar form

Given $z_1 = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ and $z_2 = 2\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.

Multiply moduli, add arguments: $z_1 z_2 = 8\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$. Divide moduli, subtract arguments: $\frac{z_1}{z_2} = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$.

13 [Intermediate] • Target: Multiplying and dividing in polar form

Given $z_1 = 5\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)$ and $z_2 = \cos\frac{\pi}{5} + i\sin\frac{\pi}{5}$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.

$z_1 z_2 = 5\left(\cos\frac{3\pi}{5} + i\sin\frac{3\pi}{5}\right)$. $\frac{z_1}{z_2} = 5\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)$.

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14 **[Intermediate]** • **Target: Multiplying and dividing in polar form**

Find the modulus and argument of $z = \frac{(1+i)^3}{1-i}$, using De Moivre's theorem for the numerator and polar division.

$1+i$ has modulus $\sqrt{2}$ and argument $\frac{\pi}{4}$, so $(1+i)^3$ has modulus $2\sqrt{2}$ and argument $\frac{3\pi}{4}$. $1-i$ has modulus $\sqrt{2}$ and argument $-\frac{\pi}{4}$. Dividing: modulus $= \frac{2\sqrt{2}}{\sqrt{2}} = 2$; argument $= \frac{3\pi}{4} - \left(-\frac{\pi}{4}\right) = \pi$. (This means $z = -2$, purely real.)

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: De Moivre's theorem** • **Cross-Lesson: 1.01 Functions: Modulus & Inverse**

Use De Moivre's theorem to find $\left[3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^4$, giving your answer in Cartesian form.

$3^4\left(\cos\frac{4\pi}{6} + i\sin\frac{4\pi}{6}\right) = 81\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 81\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -\frac{81}{2} + \frac{81\sqrt{3}}{2}i$.

16 **[Challenging]** • **Target: De Moivre's theorem** • **Cross-Lesson: 10.01 Numerical Methods**

Use De Moivre's theorem to find $(1-i)^5$, giving your answer in Cartesian form.

$|1-i| = \sqrt{2}$, $\arg(1-i) = -\frac{\pi}{4}$. $(1-i)^5 = (\sqrt{2})^5\left(\cos\left(-\frac{5\pi}{4}\right) + i\sin\left(-\frac{5\pi}{4}\right)\right)$. Since $-\frac{5\pi}{4}$ is equivalent to $\frac{3\pi}{4}$: $= 4\sqrt{2}\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right) = -4 + 4i$.

17 **[Challenging]** • **Target: Finding roots using De Moivre's theorem** • **Cross-Lesson: 11.01 Differential Equations**

Find the three cube roots of $z = 27\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$, giving arguments in $[0, 2\pi)$.

Cube roots have modulus $27^{1/3} = 3$ and arguments $\frac{\pi/2 + 2\pi k}{3}$ for $k = 0, 1, 2$: $\frac{\pi}{6}$, $\frac{5\pi}{6}$, $\frac{3\pi}{2}$. Roots: $3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$, $3\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$, $3\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$.

18 **[Challenging]** • **Target: Finding roots using De Moivre's theorem** • **Cross-Lesson: 12.01 Poisson Distribution**

Find all four fourth roots of -16 (i.e. solve $z^4 = -16$), giving arguments in $[0, 2\pi)$.

Write $-16 = 16(\cos\pi + i\sin\pi)$. Fourth roots have modulus $16^{1/4} = 2$ and arguments $\frac{\pi + 2\pi k}{4}$ for $k = 0, 1, 2, 3$: $\frac{\pi}{4}$, $\frac{3\pi}{4}$, $\frac{5\pi}{4}$, $\frac{7\pi}{4}$. Roots: $2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$, $2\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$, $2\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right)$, $2\left(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\right)$.

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19 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 13.01 Continuous Random Variables

This question has three parts. An AC circuit has impedance $Z = R + iX_L$, where $R = 8\,\Omega$ (resistance) and $X_L = 6\,\Omega$ (inductive reactance). (a) Write Z in Cartesian form, find $|Z|$, and find $\arg Z$ to the nearest degree. (b) The current is $I = \frac{V}{Z}$, where $V = 100$ volts (real). Find I in the form $a + bi$, by rationalizing the denominator. (c) Find $|I|$, to two decimal places.

(a) $Z = 8 + 6i$. $|Z| = \sqrt{64 + 36} = \sqrt{100} = 10\,\Omega$. $\arg Z = \arctan\left(\frac{6}{8}\right) \approx 37^\circ$. (b) $I = \frac{100}{8 + 6i} \cdot \frac{8 - 6i}{8 - 6i} = \frac{100(8 - 6i)}{64 + 36} = \frac{100(8 - 6i)}{100} = 8 - 6i$ amps. (c) $|I| = \sqrt{8^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10.00$ amps, consistent with $|I| = \frac{|V|}{|Z|} = \frac{100}{10} = 10$.

20 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 14.01 Sampling & Estimation

The complex number z is given by $z = \frac{4 + 3\mathrm{i}}{2 - \mathrm{i}}$. Express z in the Cartesian form $x + \mathrm{i}y$, where x and y are real numbers.

Final Answer: $1 + 2\mathrm{i}$. To express z in the form $x + \mathrm{i}y$, multiply the numerator and the denominator by the complex conjugate of the denominator. The denominator is $2 - \mathrm{i}$, so its complex conjugate is $2 + \mathrm{i}$. Multiply numerator and denominator: $z = \frac{(4 + 3\mathrm{i})(2 + \mathrm{i})}{(2 - \mathrm{i})(2 + \mathrm{i})}$ Expand the numerator: $(4 + 3\mathrm{i})(2 + \mathrm{i}) = 8 + 4\mathrm{i} + 6\mathrm{i} + 3\mathrm{i}^2 = 8 + 10\mathrm{i} - 3 = 5 + 10\mathrm{i}$ Expan.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Cartesian Form & Complex Arithmetic

Given $z = 3 - 4i$, compute $z \cdot z^*$ where z^* is the complex conjugate of z.

a. **25** ✓

b. 7

c. -7

d. $25 - 24i$

$z^* = 3 + 4i$. Then $z \cdot z^* = 3^2 - (4i)^2 = 9 - (-16) = 25$.

22 [Foundational] • Target: Cartesian Form & Complex Arithmetic

Express $\frac{5 + 2i}{2 - i}$ in the form $a + bi$.

a. $\frac{8}{5} + \frac{9}{5}i$ ✓

b. $\frac{12}{5} + \frac{9}{5}i$

c. $8 + 9i$

d. $\frac{8}{3} + \frac{9}{3}i$

Multiply numerator and denominator by $(2 + i)$: $\frac{(5 + 2i)(2 + i)}{4 + 1} = \frac{10 + 5i + 4i + 2i^2}{5} = \frac{8 + 9i}{5} = \frac{8}{5} + \frac{9}{5}i$.

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23 [Foundational] • Target: Cartesian Form & Complex Arithmetic

Find the two square roots of $-5 + 12i$.

a. $\pm(2 + 3i)$ ✓

b. $\pm(3 + 2i)$

c. $\pm(2 - 3i)$

d. $\pm(3 - 2i)$

Let $(x + iy)^2 = -5 + 12i \Rightarrow x^2 - y^2 = -5$ and $2xy = 12 \Rightarrow xy = 6$. Also $x^2 + y^2 = \sqrt{25 + 144} = 13$. Adding: $2x^2 = 8 \Rightarrow x = \pm 2$. Then $y = 6/x = \pm 3$. Roots: $\pm(2 + 3i)$.

24 [Foundational] • Target: Modulus-Argument Form

Find the modulus and principal argument of $z = -1 + i\sqrt{3}$.

a. $r = 2, \theta = \frac{2\pi}{3}$ ✓

b. $r = 2, \theta = \frac{\pi}{3}$

c. $r = 4, \theta = \frac{2\pi}{3}$

d. $r = 2, \theta = -\frac{2\pi}{3}$

Modulus $r = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{4} = 2$. Point $(-1, \sqrt{3})$ is in QII: $\theta = \pi - \arctan(\sqrt{3}) = \pi - \pi/3 = 2\pi/3$.

25 [Foundational] • Target: Modulus-Argument Form

Given $z_1 = 3(\cos(\pi/4) + i\sin(\pi/4))$ and $z_2 = 2(\cos(\pi/6) + i\sin(\pi/6))$, find $\arg(z_1 z_2)$.

a. $\frac{5\pi}{12}$ ✓

b. $\frac{\pi}{12}$

c. $\frac{\pi}{24}$

d. $\frac{7\pi}{12}$

When multiplying complex numbers, arguments add: $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) = \frac{\pi}{4} + \frac{\pi}{6} = \frac{5\pi}{12}$.

26 [Foundational] • Target: Modulus-Argument Form

Express $z = 4e^{i\pi/3}$ in Cartesian form $x + iy$.

a. $2 + 2i\sqrt{3}$ ✓

b. $2\sqrt{3} + 2i$

c. $2 - 2i\sqrt{3}$

d. $4 + 4i\sqrt{3}$

$4(\cos(\pi/3) + i\sin(\pi/3)) = 4(1/2 + i\sqrt{3}/2) = 2 + 2i\sqrt{3}$.

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27 [Foundational] • Target: Loci in the Argand Plane

Describe the locus of points represented by $|z - 3 - 4i| = 5$ in the complex plane.

a. A circle with center $(3, 4)$ and radius 5 ✓

b. A circle with center $(-3, -4)$ and radius 5

c. A circle with center $(3, 4)$ and radius 25

d. A straight line perpendicular to $3 + 4i$

$|z - (3 + 4i)| = 5$ represents all points at distance 5 from $3 + 4i$, which is a circle centered at $(3, 4)$ with radius 5.

28 [Foundational] • Target: Loci in the Argand Plane

Describe the locus of points satisfying $|z - 2| = |z - 6i|$.

a. Perpendicular bisector of the segment joining $(2, 0)$ and $(0, 6)$ ✓

b. Circle passing through $(2, 0)$ and $(0, 6)$

c. Ellipse with foci at $(2, 0)$ and $(0, 6)$

d. Line segment joining $(2, 0)$ and $(0, 6)$

The equation $|z - z_1| = |z - z_2|$ represents the perpendicular bisector of the line segment joining z_1 and z_2.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Loci in the Argand Plane

Describe the geometric locus defined by $\arg(z-2)=\frac{\pi}{4}$.

a. A half-line starting from $(2,0)$ making an angle of $\pi/4$ with the positive real axis ✓

b. A full line through $(2,0)$ with gradient 1

c. A circle of radius 2

d. A ray towards the origin

$\arg(z-z_0)=\alpha$ represents a ray (half-line) emanating from z_0 at angle α, excluding the endpoint z_0.

30 [Intermediate] • Target: Roots of Polynomials & Conjugate Pairs

Given that $z=1+2i$ is a root of $z^3-3z^2+7z-5=0$, what are the other two roots?

a. $z=1-2i$ and $z=1$ ✓

b. $z=-1-2i$ and $z=1$

c. $z=1-2i$ and $z=5$

d. $z=-1+2i$ and $z=-1$

Since coefficients are real, the conjugate $1-2i$ must also be a root. Product of roots: $(1+2i)(1-2i)=5$. Since constant term is -5, the third root is $5/5=1$.

31 [Intermediate] • Target: Roots of Polynomials & Conjugate Pairs

Find a quadratic equation with real coefficients having root $z=4-3i$.

a. $z^2-8z+25=0$ ✓

b. $z^2+8z+25=0$

c. $z^2-8z+7=0$

d. $z^2-6z+25=0$

Roots are $4\pm3i$. Sum $=8$. Product $=4^2+3^2=25$. Equation is $z^2-8z+25=0$.

32 [Intermediate] • Target: Roots of Polynomials & Conjugate Pairs

Find the least value of $|z|$ for points z satisfying $|z-5-12i|\le3$.

a. 10 ✓

b. 13

c. 16

d. 7

Center of circle is $5+12i$, distance from origin is $\sqrt{25+144}=13$. The minimum distance from origin to circle is $13-r=13-3=10$.

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33 [Intermediate] • Target: Roots of Polynomials & Conjugate Pairs

Describe the locus $|z-(-11-10i)|=3$ in the Argand diagram. Find its radius.

a. 5

b. **3** ✓

c. 6

d. 4

The locus $|z-z_0|=r$ is a circle centered at $z_0=-11-10i$ with radius $r=3$.

34 [Intermediate] • Target: Basic arithmetic with complex numbers

Describe the locus $|z-(-2+2i)|=3$ in the Argand diagram. Find its radius.

a. 4

b. **3** ✓

c. 6

d. 2

The locus $|z-z_0|=r$ is a circle centered at $z_0=-2+2i$ with radius $r=3$.

35 [Intermediate] • Target: The complex conjugate and division

Describe the locus $|z-(-6-3i)|=5$ in the Argand diagram. Find its radius.

a. 7

b. 8

c. 6

d. **5** ✓

The locus $|z-z_0|=r$ is a circle centered at $z_0=-6-3i$ with radius $r=5$.

36 [Intermediate] • Target: Modulus and the Argand diagram

Describe the locus $|z-(-8-3i)|=9$ in the Argand diagram. Find its radius.

a. 11

b. **9** ✓

c. 10

d. 12

The locus $|z-z_0|=r$ is a circle centered at $z_0=-8-3i$ with radius $r=9$.

37 [Intermediate] • Target: Solving equations with complex roots

Describe the locus $|z-(1+2i)|=10$ in the Argand diagram. Find its radius.

a. 2

b. 13

c. **10** ✓

d. 1

The locus $|z-z_0|=r$ is a circle centered at $z_0=1+2i$ with radius $r=10$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Converting to polar (modulus-argument) form • Cross-Lesson: 12.01 Poisson Distribution

Describe the locus $|z - (3 - 10i)| = 1$ in the Argand diagram. Find its radius.

a. 5

b. 3

c. **1** ✓

d. 4

The locus $|z - z_0| = r$ is a circle centered at $z_0 = 3 - 10i$ with radius $r = 1$.

39 [Challenging] • Target: Multiplying and dividing in polar form • Cross-Lesson: 13.01 Continuous Random Variables

Describe the locus $|z - (-1 - 6i)| = 1$ in the Argand diagram. Find its radius.

a. 3

b. **1** ✓

c. 4

d. 2

The locus $|z - z_0| = r$ is a circle centered at $z_0 = -1 - 6i$ with radius $r = 1$.

40 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 14.01 Sampling & Estimation

Describe the locus $|z - (1 + 5i)| = 2$ in the Argand diagram. Find its radius.

a. **2** ✓

b. 1

c. 6

d. 5

The locus $|z - z_0| = r$ is a circle centered at $z_0 = 1 + 5i$ with radius $r = 2$.

41 [Challenging] • Target: Finding roots using De Moivre's theorem • Cross-Lesson: 15.01 Hypothesis Testing

Describe the locus $|z - (7 - 6i)| = 9$ in the Argand diagram. Find its radius.

a. **9** ✓

b. 11

c. 12

d. 7

The locus $|z - z_0| = r$ is a circle centered at $z_0 = 7 - 6i$ with radius $r = 9$.

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42 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 2.01 Exponentials & Logarithms

Describe the locus $|z - (6 + 8i)| = 4$ in the Argand diagram. Find its radius.

a. 6

b. 7

c. 8

d. **4** ✓

The locus $|z - z_0| = r$ is a circle centered at $z_0 = 6 + 8i$ with radius $r = 4$.

43 [Challenging] • Target: Cartesian Form & Complex Arithmetic • Cross-Lesson: 3.01 Advanced Trigonometry

Describe the locus $|z - (3 - 6i)| = 8$ in the Argand diagram. Find its radius.

a. **8** ✓

b. 11

c. 3

d. 10

The locus $|z - z_0| = r$ is a circle centered at $z_0 = 3 - 6i$ with radius $r = 8$.

44 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 4.01 Differentiation

Describe the locus $|z - (6 + 3i)| = 2$ in the Argand diagram. Find its radius.

a. 3

b. **2** ✓

c. 5

d. 6

The locus $|z - z_0| = r$ is a circle centered at $z_0 = 6 + 3i$ with radius $r = 2$.

45 [Challenging] • Target: Loci in the Argand Plane • Cross-Lesson: 5.01 Integration

Describe the locus $|z - (2i)| = 1$ in the Argand diagram. Find its radius.

a. **1** ✓

b. 4

c. 2

d. 5

The locus $|z - z_0| = r$ is a circle centered at $z_0 = 2i$ with radius $r = 1$.

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